

DA499 Graduation Project

A Deep Learning Framework for Bone Fracture Detection and Segmentation

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Acknowledgement

"First and foremost, I extend my deepest gratitude to my parents, my source of strength. Their unwavering prayers and support made this achievement possible."

Special thanks to **Dr. Enas ALIHashashneh** for her invaluable scientific guidance, and **Dr. Abdullah Talafah** for sharing his medical expertise in orthopedics.

| Problem Statement & Motivation

The Challenge

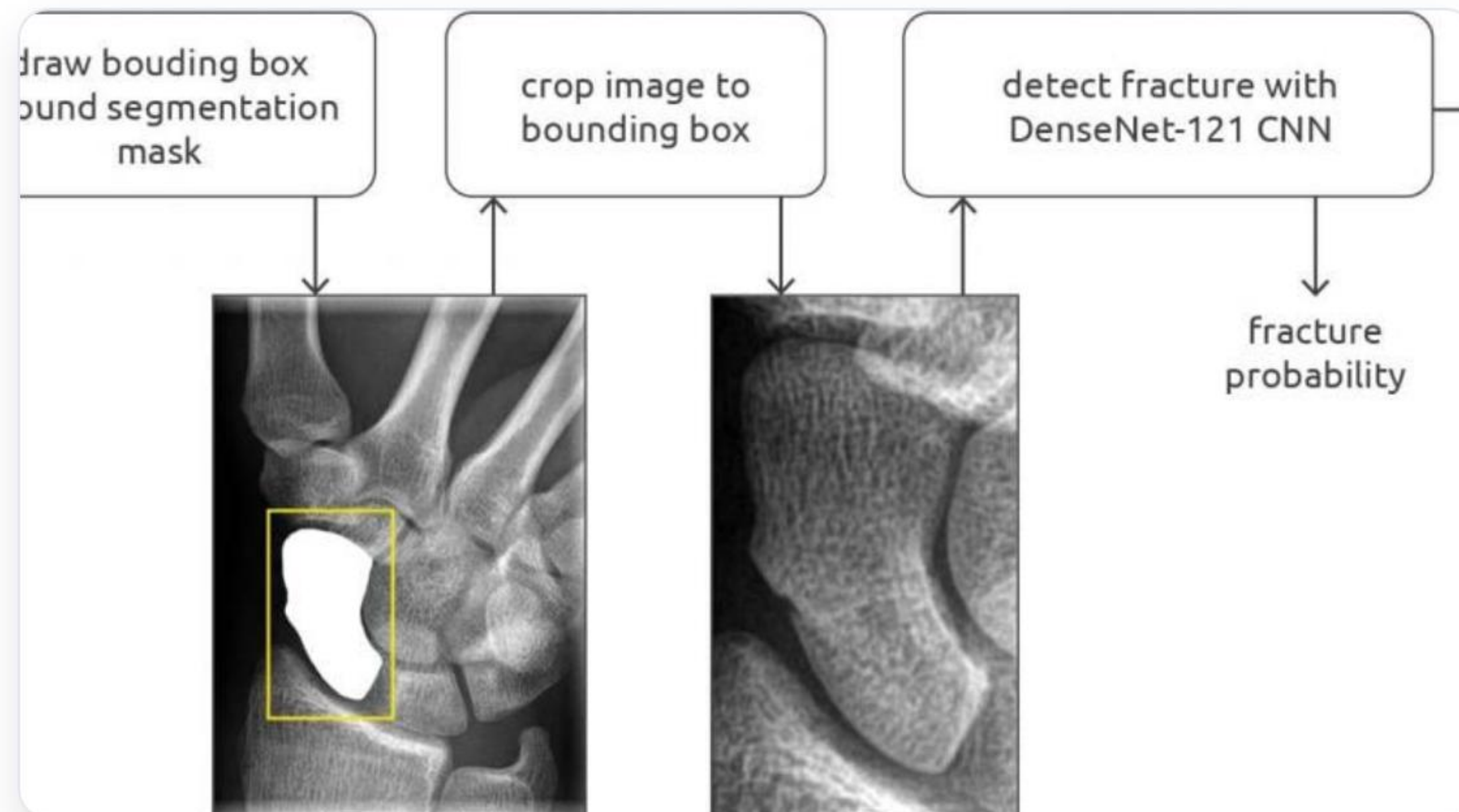
Diagnosing bone fractures from X-rays in emergency rooms is high-pressure work. Human interpretation faces significant obstacles:

- **High Workload:** Fatigue leads to missing micro- or hairline fractures.
- **Feature Loss:** Hairline fractures are only a few pixels wide; traditional AI models may lose these details during feature pooling.



Subtle details require dense connectivity.

Multi-Task Learning Approach



The Solution: FractureNet

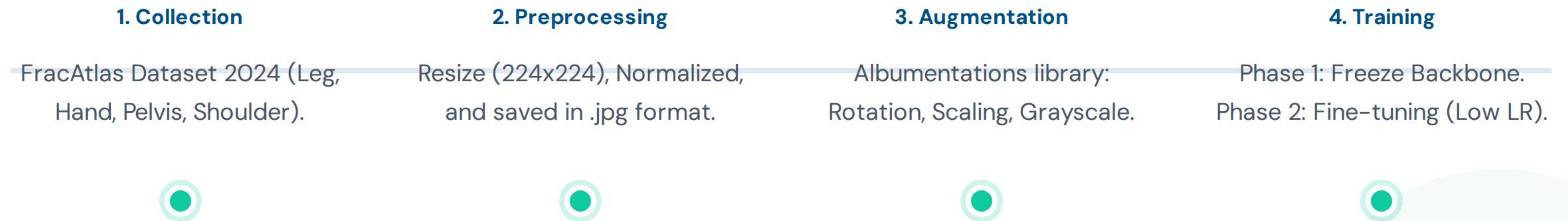
Integrated three diagnostic functions in a single pass:

1. Classification: Confirm the injury.

2. Localization: Pinpoint location via Bounding Boxes.

3. Segmentation: Precise pixel-level boundary extraction.

Methodology Workflow



| Dataset Analysis (Class Imbalance)

The FracAtlas dataset shows a significant class imbalance that required **Class Weights** and **Focal Loss** techniques to handle:

- **Fractured:** 717 Images (17.6%)
- **Non-Fractured:** 3,366 Images (82.4%)



Architecture Comparison Overview



Note: Overall Accuracy is shown. Recall varies significantly across models.

Comparative Detailed Metrics

Model Architecture	Accuracy	Precision (Fractured)	Recall (Fractured)	F1-Score
DenseNet121 (Primary)	0.86	0.57	0.75	0.65
EfficientNetB0	0.83	0.52	0.27	0.36
EfficientNetB3	0.75	0.35	0.50	0.41
EfficientNetB4	0.77	0.39	0.53	0.45
ResNet50V2	0.71	0.28	0.41	0.33

Primary Model Spotlight: DenseNet121

Why DenseNet121?

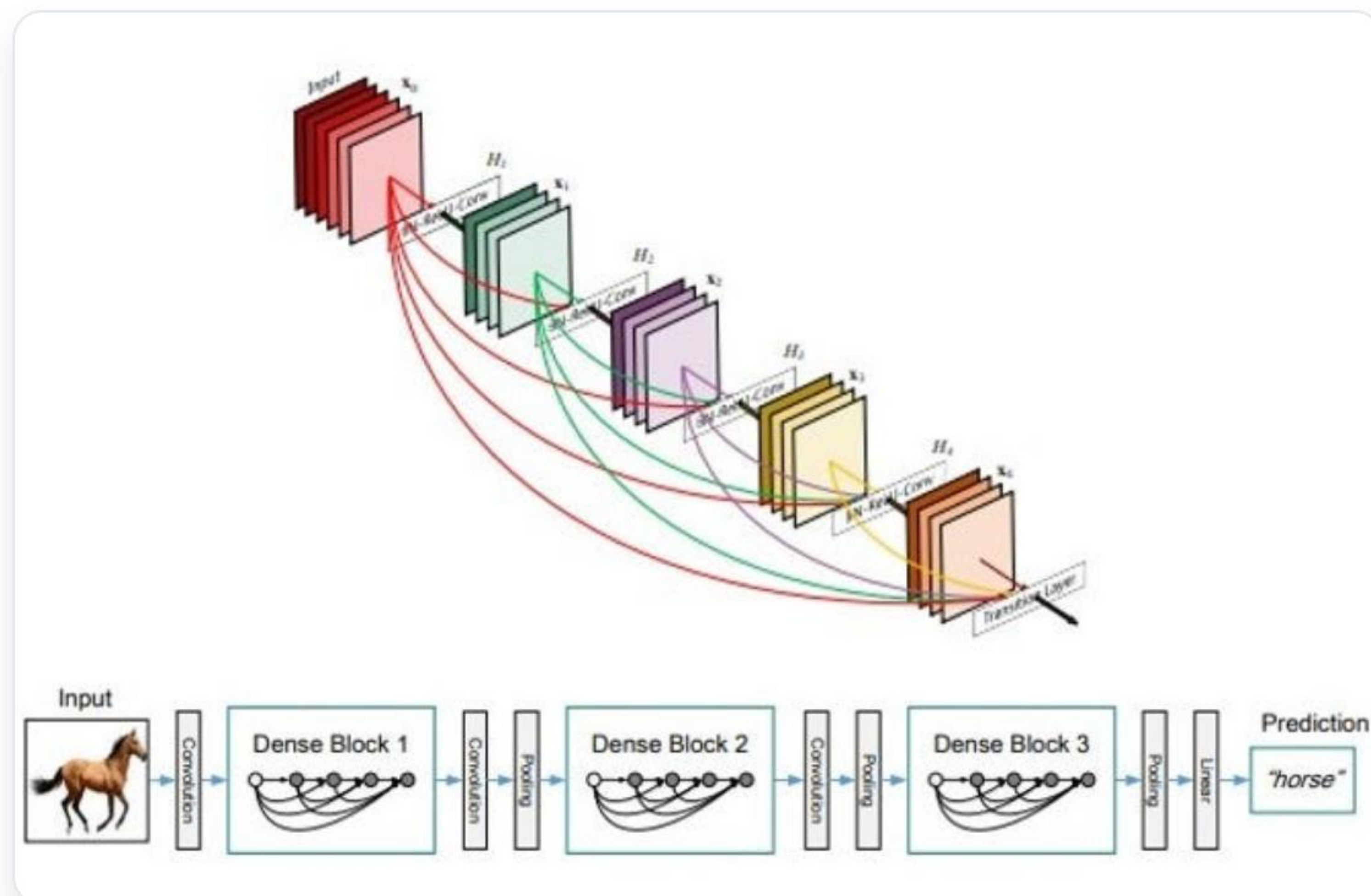
Dense links allow information to flow through layers without losing fine detail (crucial for hairline fractures).

0.92

AUC SCORE

0.86

ACCURACY



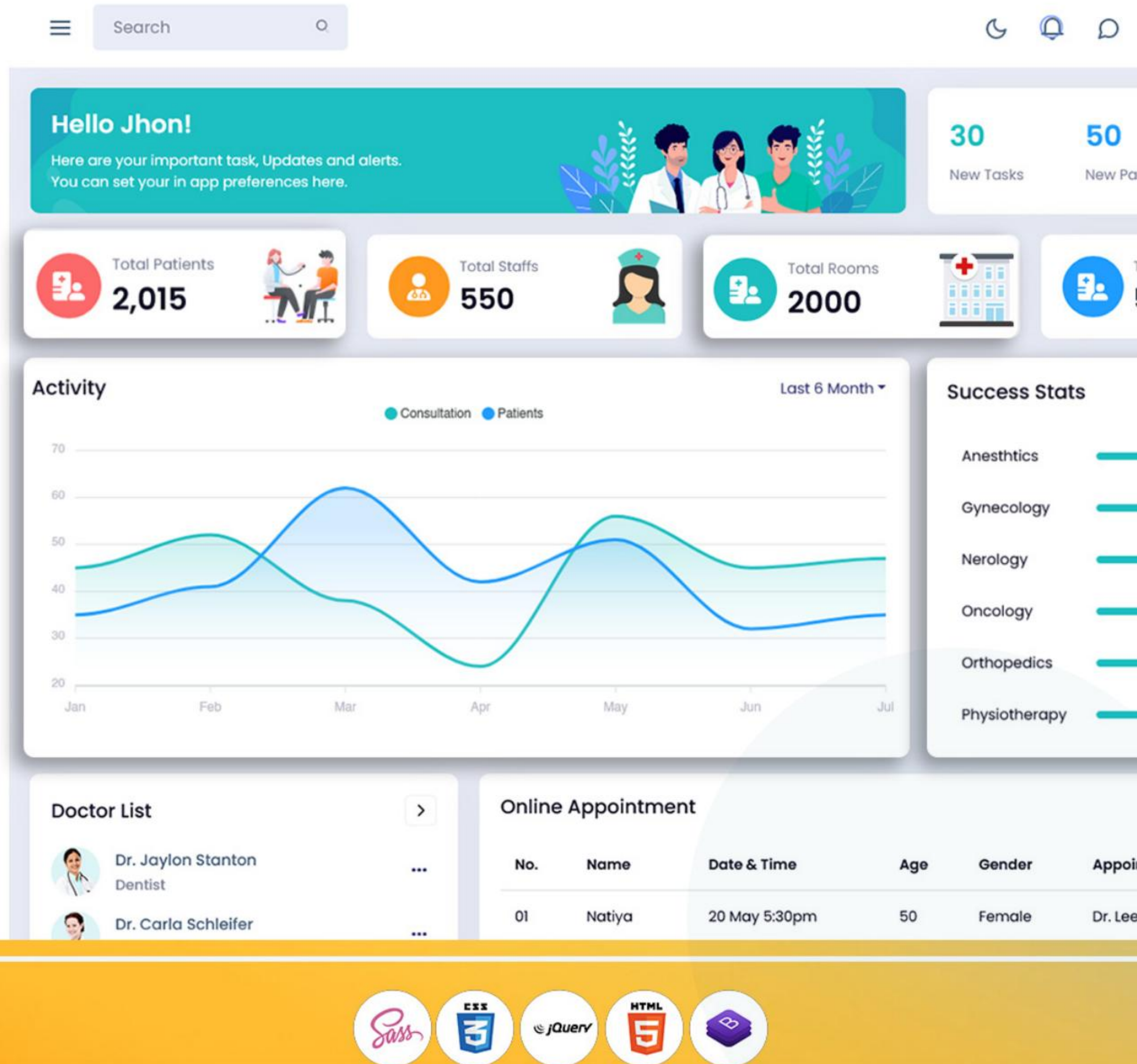
FractureNet AI System

A comprehensive, web-ready multi-tasking application built for clinical deployment:

 **Backend (FastAPI):** High-performance asynchronous processing for real-time inference.

 **Frontend:** Responsive interface featuring glass-morphism and Base64 image overlays.

Medicare Admin Dashboard Template



Conclusion & Future Directions



Key Result

DenseNet121 proved superior for capturing medical patterns. Focal loss successfully balanced a highly skewed dataset.



Limitations

Current model assumes a single fracture per image. Recall for subtle fractures remains a target for improvement (44% missed).



Future Work

Integrate Vision Transformers (ViT), process native DICOM formats, and add Grad-CAM for Explainable AI (XAI).

Thank You

Any Questions?

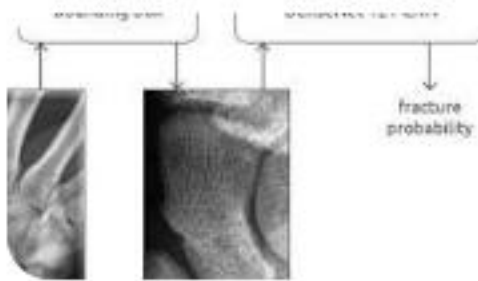
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Project: Bone Fracture Detection & Segmentation Framework

Image Sources



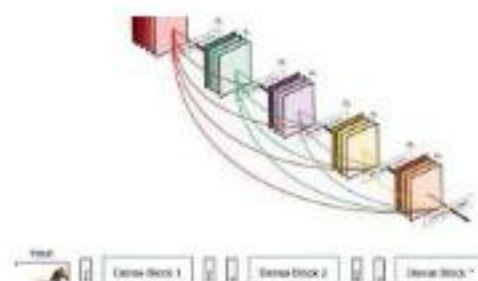
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